NUMERICAL & STATISTICAL METHODS

Subject Code : 23AS4406

UNIT - I

Solution of Algebraic & Transcendental Equations

Syllabus: Introduction - Bisection Method - Iterative method, Regula-falsi method and Newton Raphson method. System of Algebraic equations: Jacobi and Gauss Seidel method.

I. Introduction

Many problems in science & engineering lead to equations like **$f(x) = 0$**. Such equations are of two types :

- **Algebraic equation** : polynomial in x, e.g. $x^3 - 4x + 9 = 0$
- **Transcendental equation** : has trigonometric, exponential or log terms, e.g. $e^x - 3x = 0$, $x - \cos x = 0$

Exact (analytical) solution is not always possible, so we use **numerical methods** which give approximate roots by repeating a process (iteration).

Root of $f(x) = 0$: a number α such that $f(\alpha) = 0$.
Geometrically it is the point where the curve $y = f(x)$ cuts the x-axis.

Intermediate Value Property (starting point)

If $f(x)$ is continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$, then at least one real root of $f(x) = 0$ lies between a and b.

Steps to locate a root

- Make a table of $f(x)$ for $x = 0, 1, 2, 3 \ldots$ (integer values).
- Find two consecutive values where f changes sign.
- The root lies in that interval, take it as the starting interval.

Example : $f(x) = x^3 - x - 1$. $f(1) = -1$ (negative), $f(2) = 5$ (positive). So a root lies in $(1, 2)$.

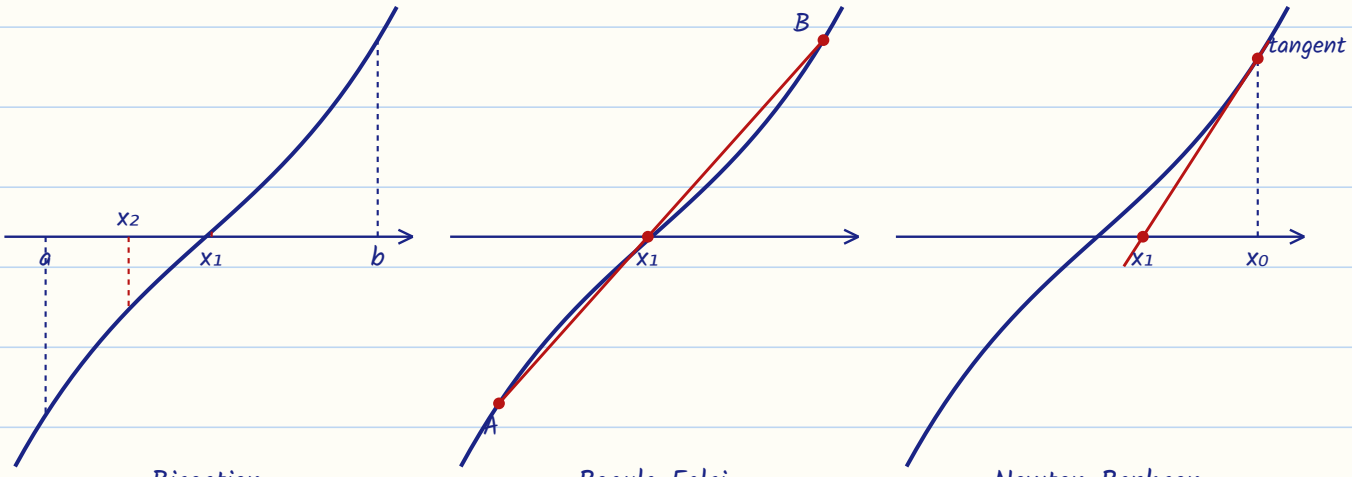
Types of methods

- **Bracketing methods** (need interval) : Bisection, Regula-Falsi
- **Open / iterative methods** (need initial guess) : Iteration, Newton-Raphson

Rate of convergence (order of a method)

If e_n = error at nth step and $e_{n+1} = C \cdot e_n^p$ then p is the **order of convergence**.

Method	Order p	Needs derivative?	Type
Bisection	1 (slow)	No	Bracketing
Regula-Falsi	1 (about 1.0)	No	Bracketing
Iteration	1 (linear)	Needs $\varphi'(x) < 1$	Open
Newton-Raphson	2 (quadratic)	Yes, $f'(x)$	Open



2. Bisection Method

Idea : repeatedly halve the interval containing the root (also called Bolzano's / interval-halving method).

Algorithm

- Choose a, b such that $f(a) \cdot f(b) < 0$.
- Find the mid-point $x_1 = (a + b) / 2$.
- If $f(x_1) = 0$, x_1 is the root. Stop.
- If $f(a) \cdot f(x_1) < 0$, root is in (a, x_1), so put $b = x_1$. Otherwise root is in (x_1, b), so put $a = x_1$.
- Repeat till $|b - a|$ is less than the required accuracy.

Each step halves the interval, so after n steps the length is $(b - a)/2^n$. Number of iterations needed for error ε : $n \geq \log(b - a) - \log \varepsilon / \log 2$

Example 1

Find a root of $x^3 - x - 1 = 0$ by bisection (correct to 2 decimals).

Here $f(1) = -1 < 0$ and $f(2) = 5 > 0$, root in $(1, 2)$.

n	a	b	x = (a+b)/2	f(x)
1	1	2	1.5	+ 0.875
2	1	1.5	1.25	− 0.2969
3	1.25	1.5	1.375	+ 0.2246
4	1.25	1.375	1.3125	− 0.0515
5	1.3125	1.375	1.34375	+ 0.0826
6	1.3125	1.34375	1.3281	+ 0.0146
7	1.3125	1.3281	1.3203	− 0.0187

Root lies between 1.3203 and 1.3281, so the root ≈ **1.32** (exact root = 1.3247).

Merits and demerits

- Merit : always converges, very simple, no derivative needed.
- Demerit : converges slowly (linear), needs many iterations, cannot find double (even multiplicity) roots.

Example 2 (to practise)

Find the positive root of $x^3 - 4x - 9 = 0$ using bisection. Take $f(2) = -9$, $f(3) = 6$. First approximation $x_1 = 2.5$, $f(2.5) = -3.375$ (negative) so root in $(2.5, 3)$. Continue ... root ≈ 2.706.

3. Method of Iteration (Fixed Point)

Write $f(x) = 0$ in the form $x = \varphi(x)$. Start with x_0 and generate

$$x_{n+1} = \varphi(x_n), \quad n = 0, 1, 2, \ldots$$

Condition for convergence : $|\varphi'(x)| < 1$ for all x in the interval containing the root. Smaller the value of $|\varphi'|$, faster the convergence.

Procedure

- Locate the root in (a, b) by sign change.
- Rewrite $f(x) = 0$ as $x = \varphi(x)$ such that $|\varphi'(x)| < 1$ in (a, b).
- Take $x_0 = a$ or b or mid-point, find $x_1 = \varphi(x_0)$, $x_2 = \varphi(x_1)$ ... till two successive values agree to required decimals.

Example 3

Find the root of $\cos x - 3x + 1 = 0$ by iteration (correct to 3 decimals).

$f(0) = 2 > 0$, $f(1) = 0.54 - 2 = -1.46 < 0$, so root in $(0, 1)$.

Write $x = (1 + \cos x)/3 = \varphi(x)$. Then $\varphi'(x) = -\sin x / 3$, $|\varphi'| \leq 1/3 < 1$, so the method converges.

Take $x_0 = 0.5$ (radians).

n	x_n	$x_{n+1} = (1 + \cos x_n)/3$
0	0.5	0.62586
1	0.62586	0.60349
2	0.60349	0.60779
3	0.60779	0.60697
4	0.60697	0.60713
5	0.60713	0.60710

Hence the root is $x \approx 0.607$ (correct to 3 decimals).

Note

If $|\varphi'(x)| > 1$ the iterates move away from root, so choose a different form of $\varphi(x)$. For $x^3 + x^2 - 1 = 0$ one may use $x = 1/\sqrt{(1 + x)}$ (since $|\varphi'| < 1$ near the root $x \approx 0.7549$).

4. Regula-Falsi Method (False Position)

Idea : replace the curve by a chord (straight line) joining the points $A(a, f(a))$ and $B(b, f(b))$. The point where the chord cuts the x-axis is the next approximation.

Equation of chord AB : $(y - f(a)) / (f(b) - f(a)) = (x - a)/(b - a)$. Put $y = 0$:

$$x_1 = [\, a \cdot f(b) - b \cdot f(a) \,] / [\, f(b) - f(a) \,]$$

Algorithm

- Take a, b with $f(a) \cdot f(b) < 0$.
- Compute x_1 by the formula above.

- If $f(a) \cdot f(x_1) < 0$ then $b = x_1$, else $a = x_1$.
- Repeat with the new interval till required accuracy.

Example 4

Find a real root of $x^3 - 2x - 5 = 0$ by Regula-Falsi.

$f(2) = -1$ (negative), $f(3) = 16$ (positive). Root in (2, 3). $a = 2$, $b = 3$.

Step 1 : $x_1 = [2(16) - 3(-1)] / [16 + 1] = 35/17 = 2.05882$, $f(x_1) = -0.3908$ (negative) so $a = 2.05882$.

Step 2 : $x_2 = 2.08126$, $f(x_2) = -0.1472$ (negative), $a = 2.08126$.

Step 3 : $x_3 = 2.08964$, $f = -0.0547$.

Step 4 : $x_4 = 2.09274$, $f = -0.0202$.

Step 5 : $x_5 = 2.09388$, $f = -0.0075$.

Values are slowly approaching the root **2.0946**. (Correct to 2 decimals : 2.09)

Comparison

- Bisection uses only the mid-point; Regula-Falsi uses the function values so it is usually faster.
- But one end point often stays fixed, so the convergence is only linear.

5. Newton-Raphson Method

Idea : replace the curve near the root by its tangent at x_0. Tangent at $(x_0, f(x_0))$: $y - f(x_0) = f'(x_0)(x - x_0)$. It cuts x-axis ($y = 0$)

at x_1.

$$x_{n+1} = x_n - f(x_n) / f'(x_n) \ , \ n = 0, 1, 2 \ ...$$

Algorithm

- Choose initial guess x_0 close to root (usually the end where $f(x) \cdot f''(x) > 0$).
- Compute $f(x_n)$ and $f'(x_n)$, find x_{n+1}.
- Stop when $|x_{n+1} - x_n| < \varepsilon$.

Convergence condition : $|f(x) \cdot f''(x)| < |f'(x)|^2$ in the neighbourhood of root. Order of convergence is **2** (quadratic), i.e. number of correct decimal places nearly doubles every step.

Example 5 : square root formula

Find $\sqrt{12}$: this is the root of $x^2 - 12 = 0$. Here $f' = 2x$ so

$x_{n+1} = x_n - (x_n^2 - 12)/(2x_n) = \frac{1}{2} (x_n + 12/x_n)$

Take $x_0 = 3$: $x_1 = \frac{1}{2}(3 + 4) = 3.5$; $x_2 = \frac{1}{2}(3.5 + 3.4286) = 3.46429$; $x_3 = 3.46410$. So $\sqrt{12} =$ **3.4641**.

Example 6

Find the root of $x^3 - 2x - 5 = 0$ near 2 by Newton-Raphson.

$f(x) = x^3 - 2x - 5$, $f'(x) = 3x^2 - 2$. $x_0 = 2$:

$x_1 = 2 - (-1)/10 = 2.1$

$x_2 = 2.1 - f(2.1)/f'(2.1) = 2.1 - (0.061/11.23) = 2.09457$

$x_3 = 2.09455$, so root = **2.0946**

Compare : Regula-Falsi needed more than 5 steps to reach the same accuracy.

Special formulae from Newton-Raphson

To find	Formula (x_{n+1})
Reciprocal 1/N	$x_n(2 - N\,x_n)$
Square root $\sqrt{N}$	$\frac{1}{2}\,(\,x_n + N/x_n\,)$
kth root of N	$[\,(k-1)\,x_n + N/x_n^{(k-1)}\,]\,/\,k$
Reciprocal sq. root $1/\sqrt{N}$	$\frac{1}{2}\,x_n\,(3 - N\,x_n^2)$

Demerits

- Fails if $f'(x) = 0$ near the root; needs derivative every step; wrong x_0 may diverge.

6. System of Linear Equations : Iterative Methods

Direct methods (Gauss elimination) give exact answers after fixed steps. For large systems with many zeros, **iterative methods** are better. Let the system be

$a_1x + b_1y + c_1z = d_1$, $a_2x + b_2y + c_2z = d_2$, $a_3x + b_3y + c_3z = d_3$

Condition (diagonally dominant) : in every equation the coefficient of the leading unknown is larger in magnitude than the sum of the other coefficients : $|a_1| > |b_1| + |c_1|$, $|b_2| > |a_2| + |c_2|$, $|c_3| > |a_3| + |b_3|$. This guarantees convergence.

Rewrite as : $x = (d_1 - b_1y - c_1z)/a_1$, $y = (d_2 - a_2x - c_2z)/b_2$, $z = (d_3 - a_3x - b_3y)/c_3$

6.1 Jacobi's Method

Start with $x_0 = y_0 = z_0 = 0$ (initial values). Compute all new values **using only old values** of previous iteration:

$$x^{(k+1)} = (d_1 - b_1y^{(k)} - c_1z^{(k)})/a_1 \text{ , similarly } y, z$$

6.2 Gauss-Seidel Method

Same equations, but **use the latest available value immediately** : as soon as $x^{(k+1)}$ is found, it is used in the y-equation, and $x^{(k+1)}$, $y^{(k+1)}$ in the z-equation.

Example 7

Solve $10x + y + z = 12$, $2x + 10y + z = 13$, $2x + 2y + 10z = 14$.

Diagonally dominant ($10 > 2$, $10 > 3$, $10 > 4$). So $x = (12 - y - z)/10$, $y = (13 - 2x - z)/10$, $z = (14 - 2x - 2y)/10$.

Jacobi (start 0, 0, 0)

Iteration	x	y	z
1	1.2	1.3	1.4
2	0.93	0.92	0.90
3	1.018	1.024	1.030
4	0.9946	0.9934	0.9916

Tends to $x = 1$, $y = 1$, $z = 1$.

Gauss-Seidel (start 0, 0, 0)

Iteration	x	y	z
1	1.2	1.06	0.948
2	0.9992	1.0054	0.9991
3	0.9996	1.0002	1.0001
4	1.0000	1.0000	1.0000

Gauss-Seidel reaches the answer in about half the iterations.

Gauss-Seidel converges **about twice as fast** as Jacobi (when both converge) and needs less storage since new values overwrite old ones.

Jacobi vs Gauss-Seidel

Jacobi	Gauss-Seidel
Uses old values only	Uses latest values
Slower convergence	Faster (twice)
Needs 2 sets of storage	One set only
Also called method of simultaneous displacement	Method of successive displacements

7. Important Points & Quick Revision

- Bisection : $x_{n+1} = (a + b)/2$, always converges, order 1.
- Iteration : $x = \varphi(x)$, needs $|\varphi'(x)| < 1$, order 1.
- Regula-Falsi : $x = [a\ f(b) - b\ f(a)] / [f(b) - f(a)]$, order ≈ 1.
- Newton-Raphson : $x = x - f/f'$, order 2, fastest.
- Jacobi / Gauss-Seidel : need diagonal dominance, G-S twice faster.

Typical exam questions

- Find a real root of $x^3 - 4x - 9 = 0$ by bisection (answer 2.706).
- Find the root of $x\,e^x = 2$ by Newton-Raphson / Regula-Falsi (root ≈ 0.8526).
- Solve $27x + 6y - z = 85$, $6x + 15y + 2z = 72$, $x + y + 54z = 110$ by Gauss-Seidel ($x = 2.43$, $y = 3.57$, $z = 1.92$).
- Derive Newton-Raphson formula. Find $\sqrt{N}$ and $1/N$ using it.